

Minimum and Maximum Mode of 8086

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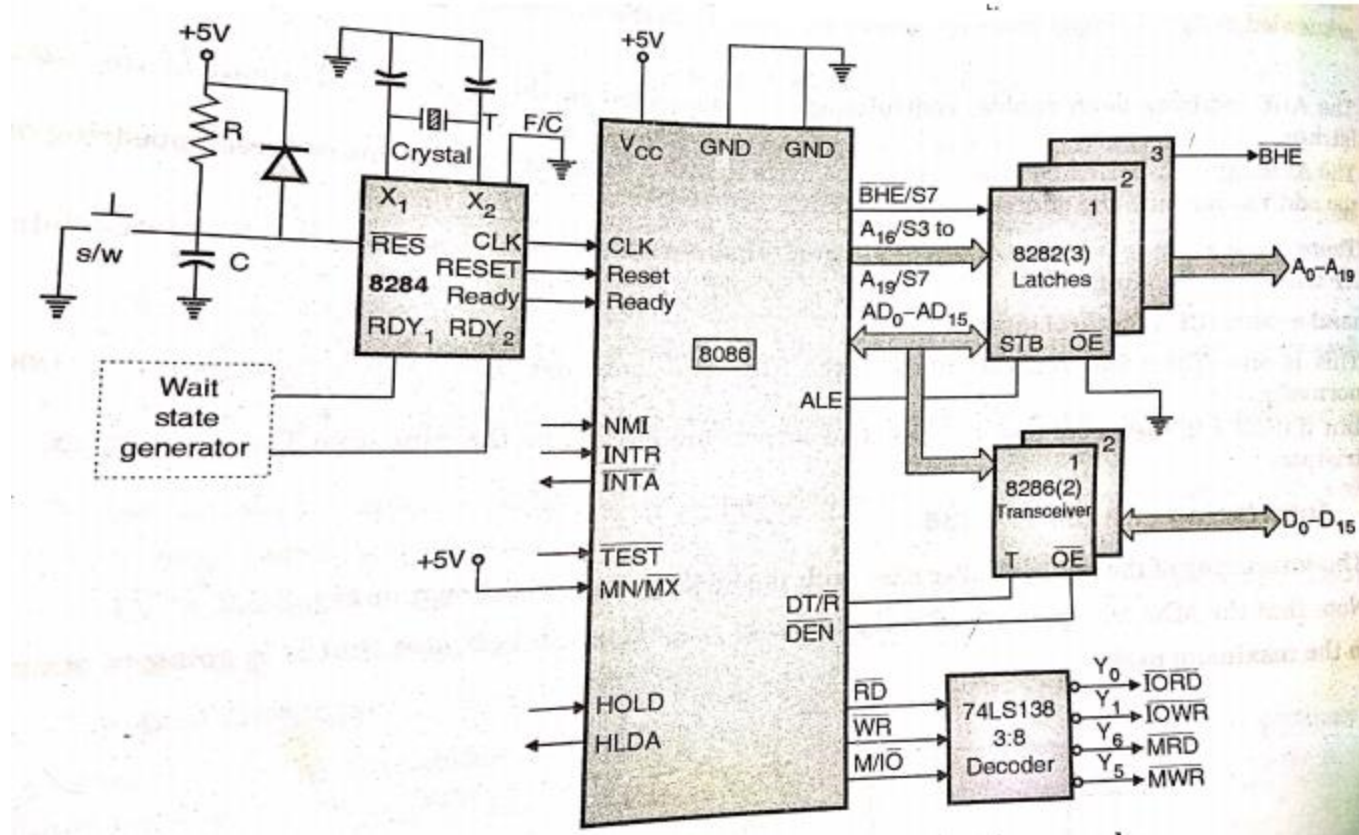
Minimum and maximum Moode

- Minimum Mode : Uniprocessor Mode
- Maximum Mode : Multiple processor mode

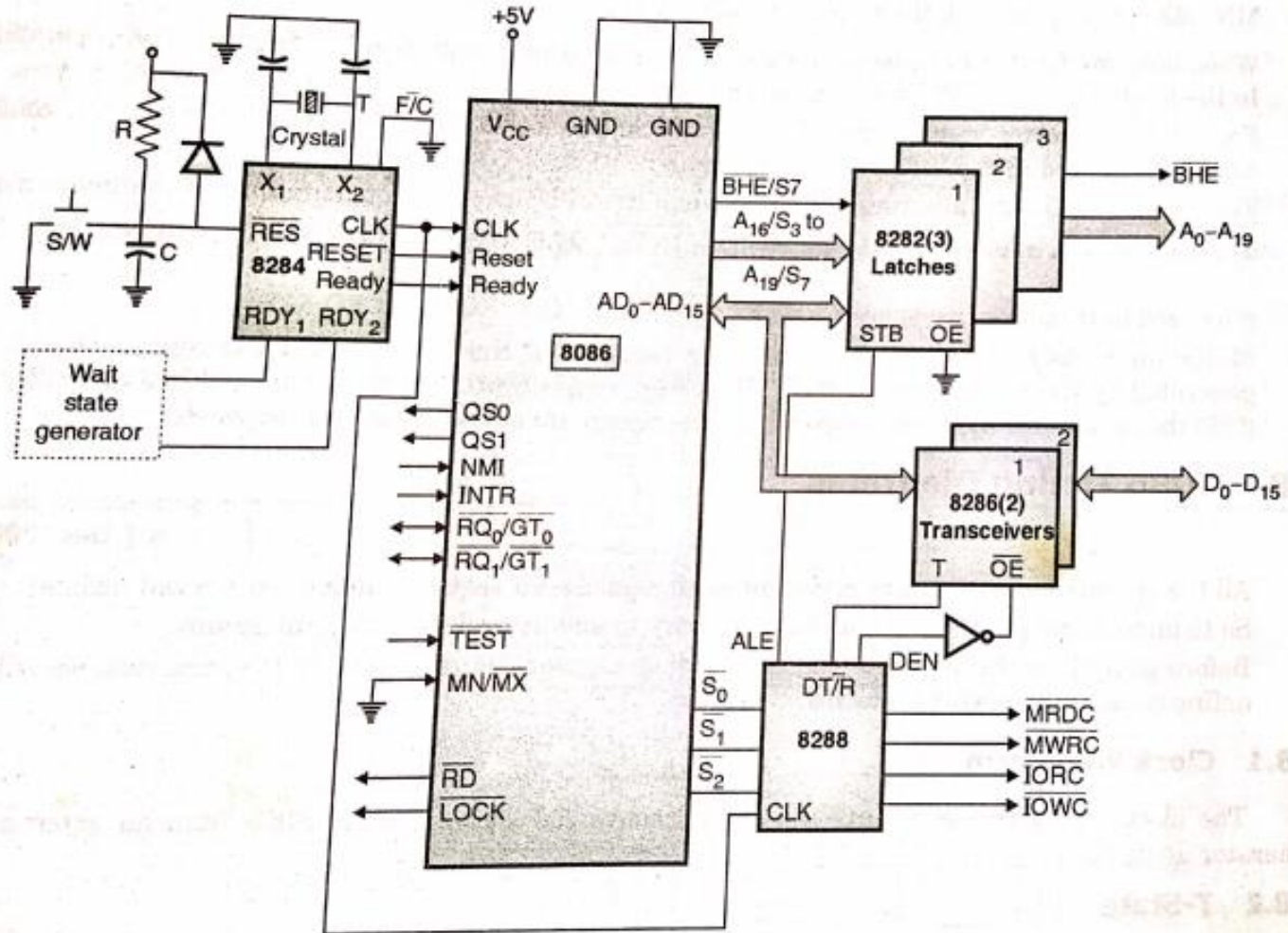
Some Common Components of 8086 system are:

- 8284- Clock Generator and driver
- 8282/8283 – Octal Latch
- 8286 – Octal bus Transceiver
- 74138- Decoder
- 8288- Bus Controller

Minimum Mode -8086

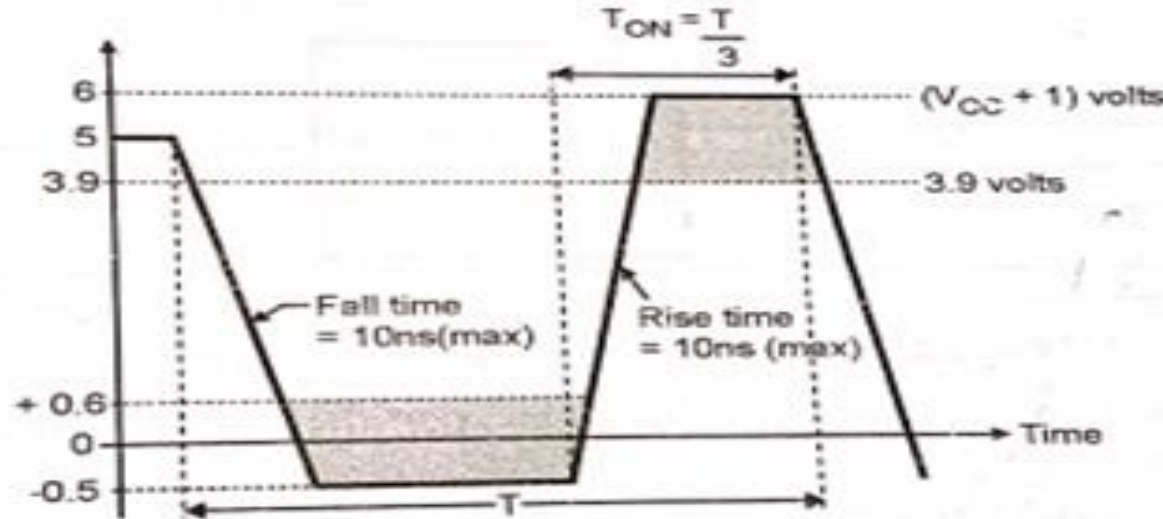


Maximum Mode- 8086



8284-Clock Generator & driver

Clock requirement of 8086:



Clock signal required by 8086/8088

Specifications of clock signal

1.	Rise and fall times	:	10 nsec. (maximum)
2.	Maximum clock frequency	:	8 MHz for 8086 – 2
		:	5 MHz for 8086
		:	10 MHz for 8086 – 1
3.	Minimum clock frequency	:	2 MHz
4.	Duty cycle = $(T_{on} / T) \times 100$	:	33%
5.	Low voltage range	:	- 0.5 V to + 0.6 V
6.	High voltage range	:	3.9 V to $(V_{cc} + 1)$ V.

6.0 MHz has been decided by keeping the refreshi

8284- Pin diagram

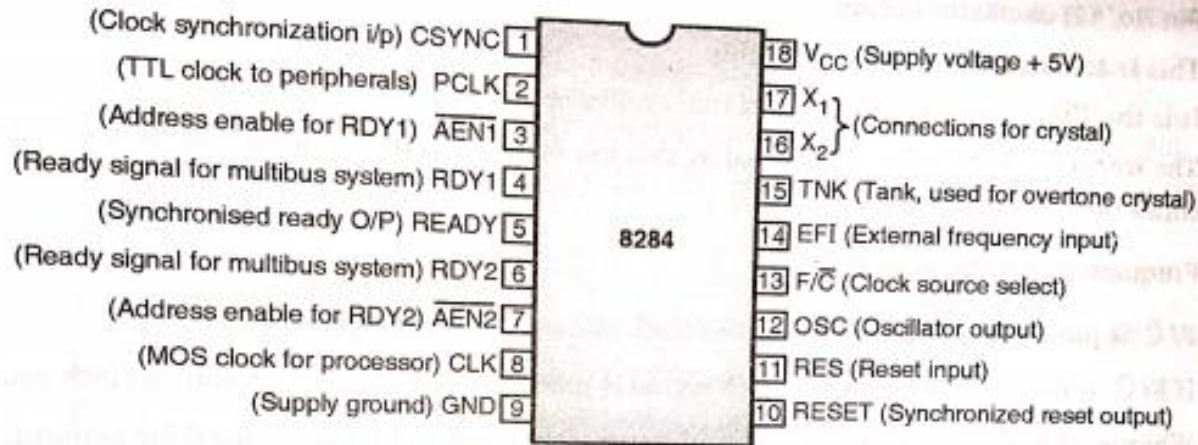


Fig. 3.2.2(b) : Pin configuration and explanation for 8284

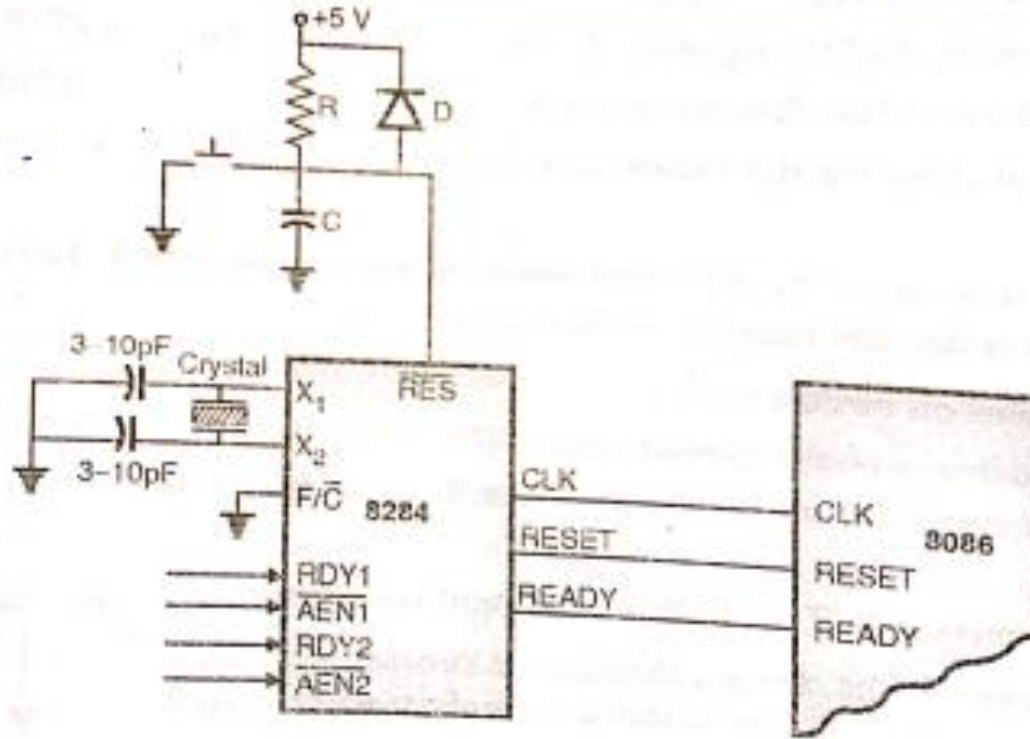
3.2.2 Features of 8284

- Generates system clock signal for 8086, 8088 processors and the peripherals.
- Operates on a single + 5V supply.
- Available in the 18 pin package.
- Can use either a crystal or a TTL external signal as frequency source.
- Generates the system reset output.
- Provides the synchronication between local ready and microprocessor ready signals.
- Capable of clock synchronization with other 8284 chips.

Interfacing 8284 with 8086

Interfacing 8284 to 8086

The typical connections for 8284 and its interfacing with the 8086 processor is shown in Fig.



Cntd.. Interfacing 8284 with 8086

RESET

- Pin no. 10 is the reset output pin. It is an active high pin which is used to reset the 8086 family processors.
- The timing characteristics of the Reset signal on this pin are decided by the external circuitry connected to the $\overline{\text{RES}}$ input.
- The RESET pin is connected directly to the RESET input of 8086 (see interfacing of 8284 with 8086 in section 3.2.4).

READY (Pin 5)

- READY (pin 5) is an active high output signal generated by 8284. This pin is connected directly to the READY input of the 8086 processor as shown in Fig. 3.2.2(b).
- This output signal is synchronized with the two Ready input signals (RDY 1 and RDY 2) of 8284.

RDY 1 and RDY 2

- RDY 1 and RDY 2 are the input signals available from the external devices on pins 4 and 6 respectively.
- Whenever an external device wants to introduce a "wait" T-state for proper data transfer, it will activate (force to 1) the RDY 1 or RDY 2 input.
- A synchronized READY signal will be produced by 8284 and applied to 8086.

Octal Latch – 8282/8283

Fig. 3.2.4 : Interfacing 8282 to 8086

8282 / 8283 : An Octal Latch

"A latch is a register with a specific purpose. This group of D flip-flops is used to hold values to be output to other devices."

For example, if you wanted to control a number of LEDs say for a level meter, a latch would hold the appropriate 1's and 0's to drive those LEDs while the processor went off to do other things.

Before going into the details of 8282/8283 octal latch let us see its utility and why do we need to use the latch.

Address data bus of 8086

- The address data bus of 8086 is a **multiplexed** bus. That means the same pins (AD_0 to AD_{15}) are being used as address lines as well as data lines.
- The same lines are being used to carry the address information as well as data, on the time sharing basis, and hence are to be demultiplexed.
- The remaining four address lines A_{16} to A_{19} also are multiplexed with the S_3 to S_6 signals (A_{16} / S_3 to A_{19} / S_6), and hence are to be demultiplexed.
- Also the $\overline{BHE} / S_7$ signal is to be demultiplexed.
- The 8086 first outputs the 20 bit address on the multiplexed lines. The presence of address on the multiplexed bus is indicated by forcing the ALE (address latch enable) pin of 8086 to HIGH.
- This address should be latched into external latches. The ALE signal is used to enable the external latches, to store the address.

- The ALE signal then goes low (0), before the address disappears from the multiplexed bus as shown in Fig. 3.3.1. This will disable the external latches.
- The address which is stored by the external latches can now be used as address bus.
- The processor will output the data on the multiplexed bus, after ALE goes low.

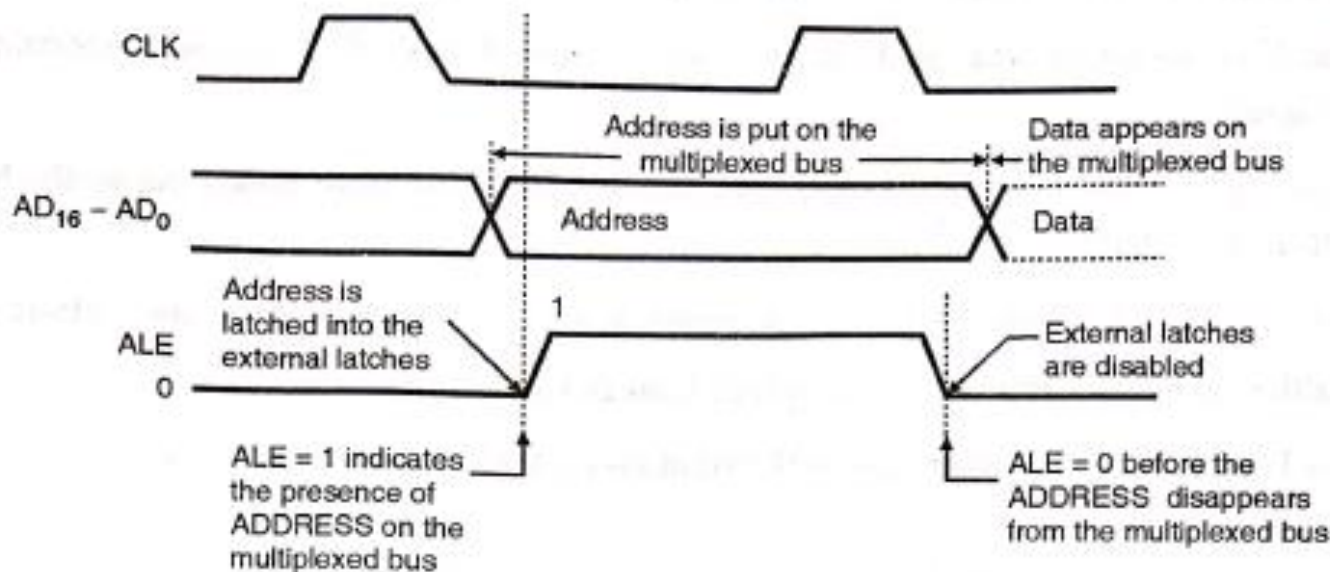


Fig. 3.3.1 : Timing diagram showing the role of ALE in the multiplexed address data bus

- The stable address required for the system memory and peripherals for the duration of bus cycle is provided by the external latches at their outputs.
- The most popularly used latches are :
 1. 8282 : Non inverting octal latch.
 2. 8283 : Inverting octal latch.

Fig. 10.10 shows diagrammatically the use of external latches to latch the address.

The ALE signal is being used as the strobe (STB) input for the latches, which enables the latches and loads the address into the latches.

Since each latch is an octal i.e. 8-bit latch, we have to use three such latches in order to latch a 20 bit address.

On receiving the HIGH (1) ALE signal from the processor, the latches will be enabled and the A_0 to A_{19} address is latched.

Before the address disappears from the multiplexed bus, the ALE reduces to 0 and the latches are disabled. So they do not respond to the data appearing on the multiplexed bus.

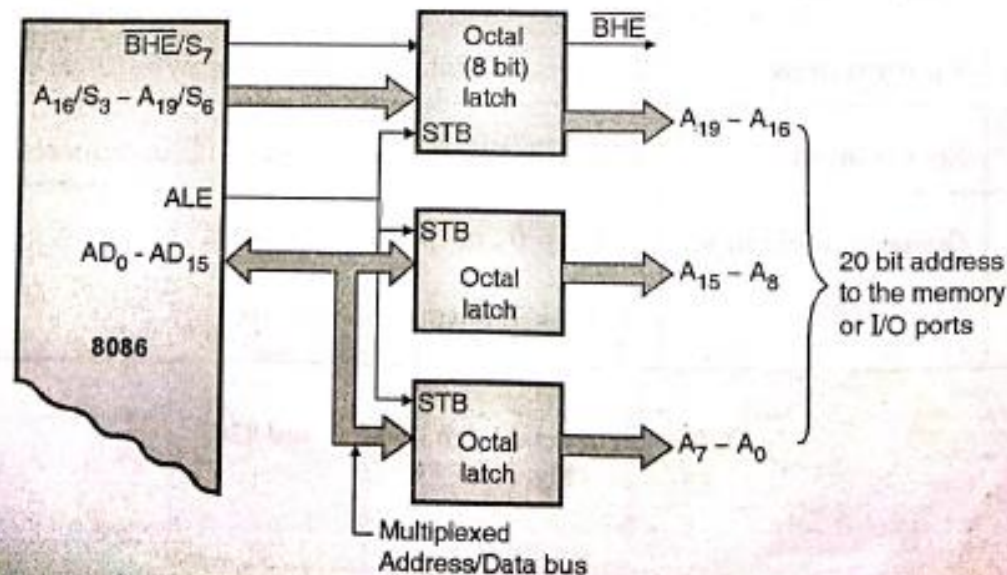
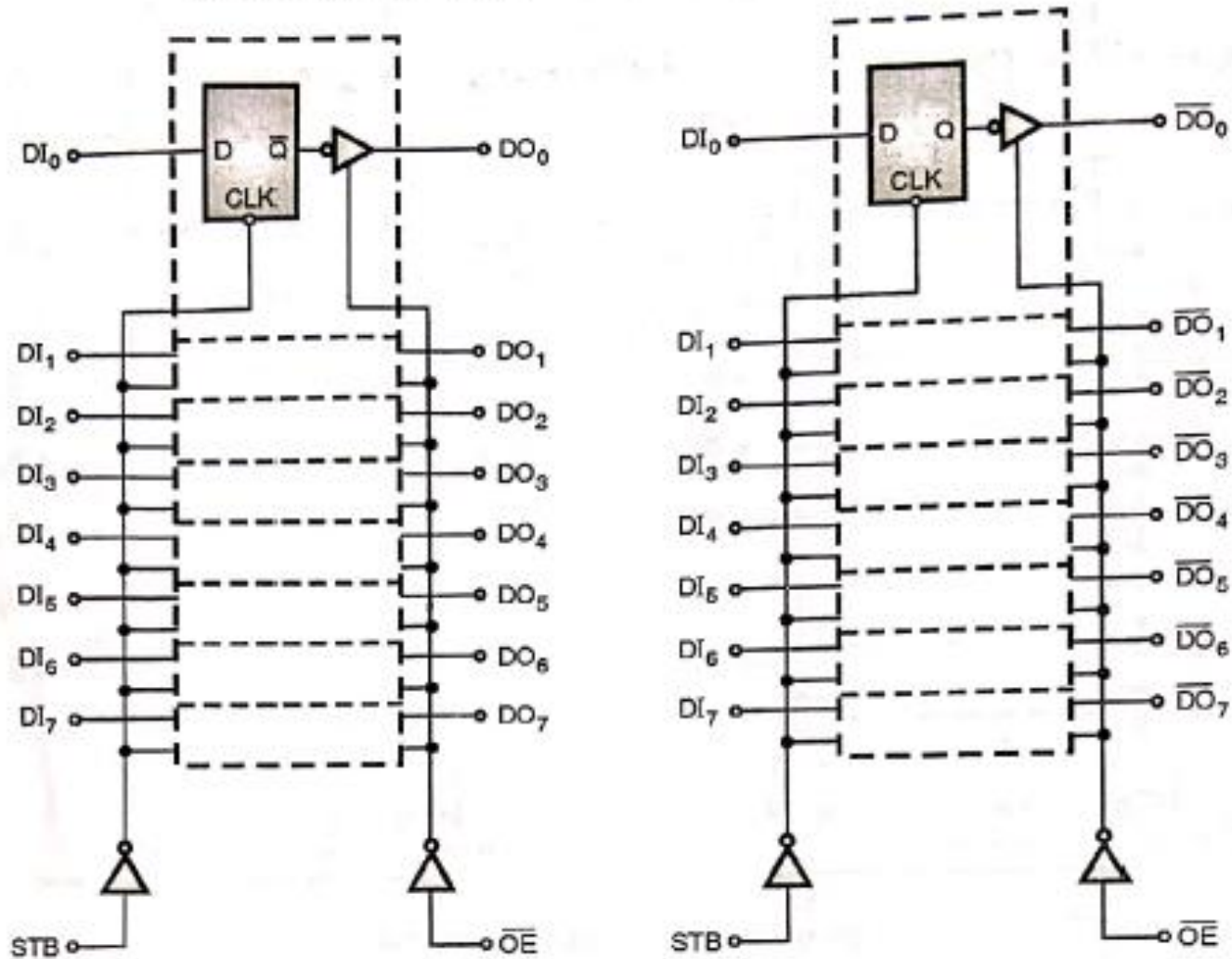


Fig. 10.10: Use of external latches for address latching

Internal Diagram of 8282/8283

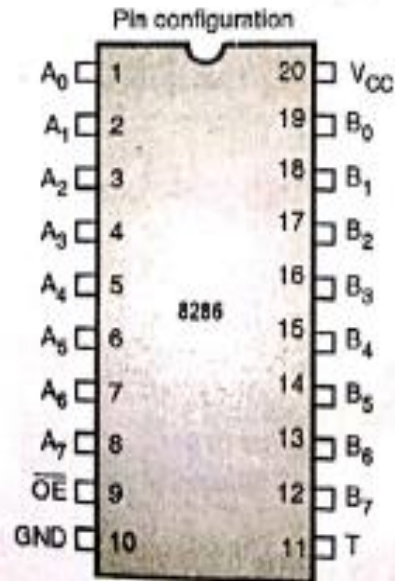
internal circuit diagram of 8282 and 8383



Bus Transceiver

8286 : Octal Bus Transceiver

- The 8286 is an octal bus transceiver which is used for driving the 16 bit data bus from the multiplexed address data bus of 8086 processor.
- The pin diagram of 8286 is shown in Fig. 3.4.1.



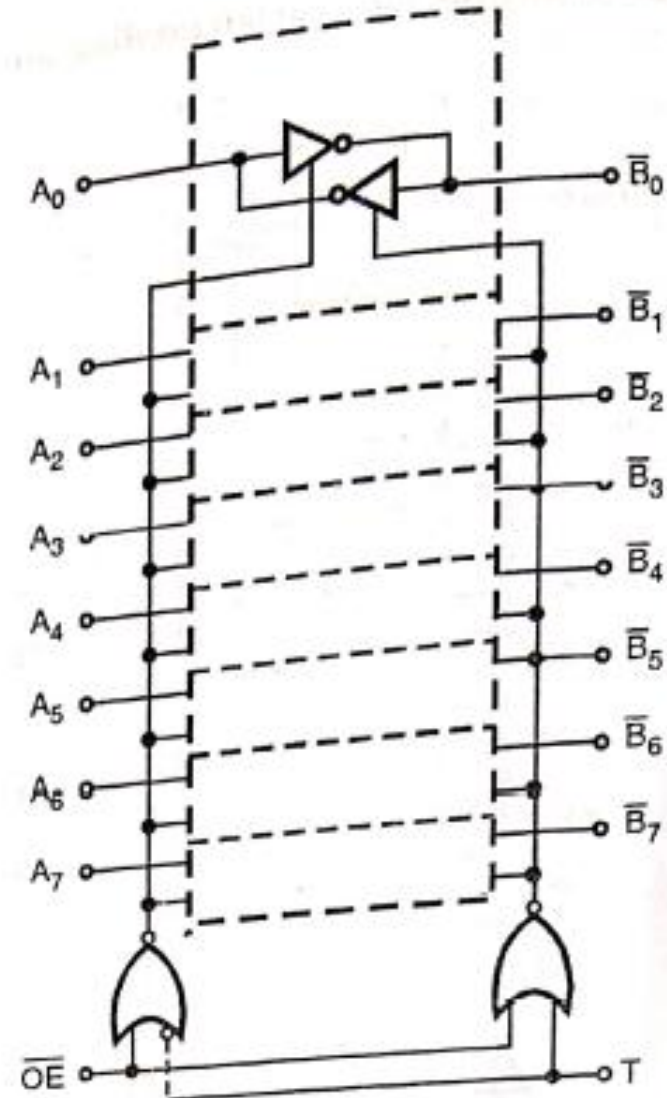
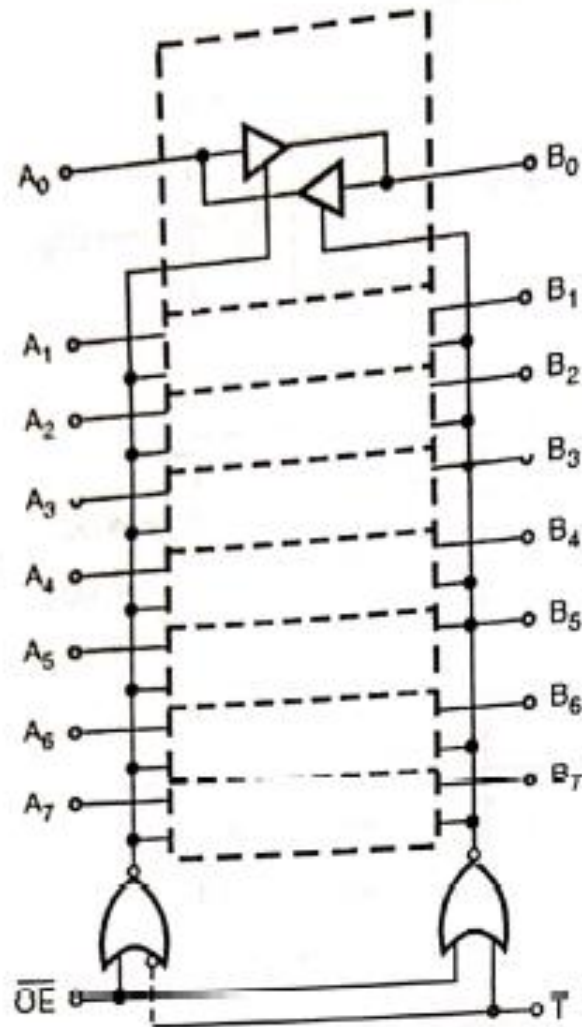
(a) 8286 pin configuration

Pin names

$A_0 - A_7$	Local bus data
$B_0 - B_7$	System bus data
$\overline{OE}$	Output enable
T	Transmit

(b) Pin names

Fig. 3.4.1



8286: Pin description

Description

1. $A_0 - A_7$
 - These 8-lines are called as the local bus data lines. They can act as input lines or as output lines depending on the status of the transmit (T) pin.
 - If $T = 1$, then $A_0 - A_7$ act as inputs, whereas if $T = 0$, then $A_0 - A_7$ will act as output lines.
2. $B_0 - B_7$
 - These 8-lines are called as system bus data lines. They too can act as input or output lines depending on the status of the transmit (T) pin.
 - If $T = 1$, then $B_0 - B_7$ will act as output lines whereas with $T = 0$, they will act as the input lines.
3. T (Transmit)
 - Pin 11 i.e. T or transmit pin is an input pin of 8286.
 - It controls the direction of the transceivers, as follows :
 If $T = 1$; $A_0 - A_7$ act as inputs and $B_0 - B_7$ act as outputs.
 If $T = 0$; $A_0 - A_7$ act as outputs with $B_0 - B_7$ acting as inputs.
4. Output Enable ($\overline{OE}$)
 - This is an active low input to 8286. It is used for enabling the appropriate driver to put his data onto the respective bus.
 - If $\overline{OE} = 1$ i.e. in active, then the outputs will be in the high impedance (tri-state) T-state.

Interfacing 8286 with 8086

The multiplexed address data bus AD_0 to AD_{16} is applied at the A inputs (A_0 to A_7) of the two 8286 ICs ($AD_0 - AD_7$ to first and $AD_8 - AD_{16}$ to the second).

The B outputs of the two 8286 chips act as the demultiplexed data bus.

The 8286 transceivers allow bi-directional data transfer. The T (transmit) input of both 8286 chips is connected to the $DT/\bar{R}$ output of 8086.

So when $DT/\bar{R} = 0$, $T = 0$, so B lines act as input lines, a lines act as the output lines and the data will flow from memory or ports to 8086, that means the read operation will take place.

On the other hand if $DT/\bar{R} = 1$, then $T = 1$ and the data will get transferred from the processor to the memory or ports.

The $\overline{OE}$ lines of both the 8286 chips are connected to the $\overline{DEN}$ output of 8086. This signal is used to enable or disable the transceivers. With $\overline{DEN} = 0$, the transceivers are enabled and with $\overline{DEN} = 1$ they are disabled and go into tri-state.

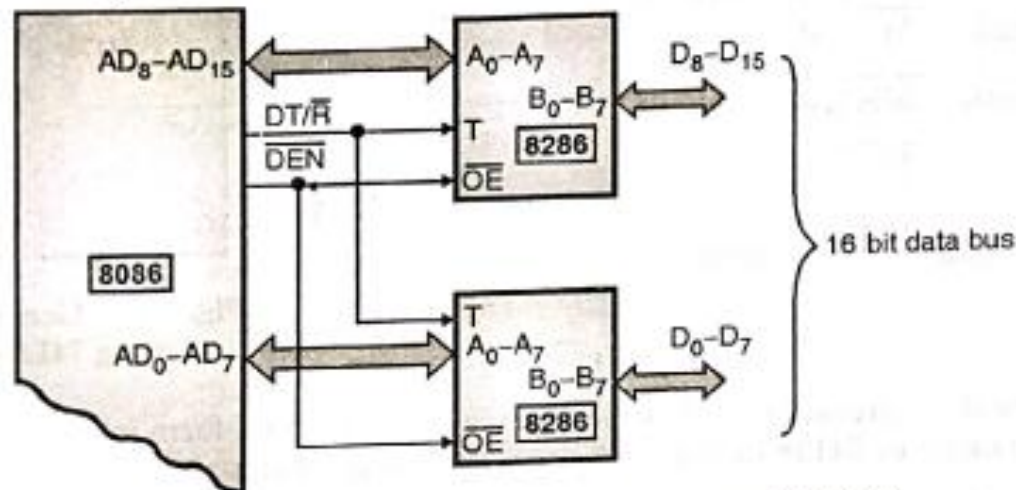
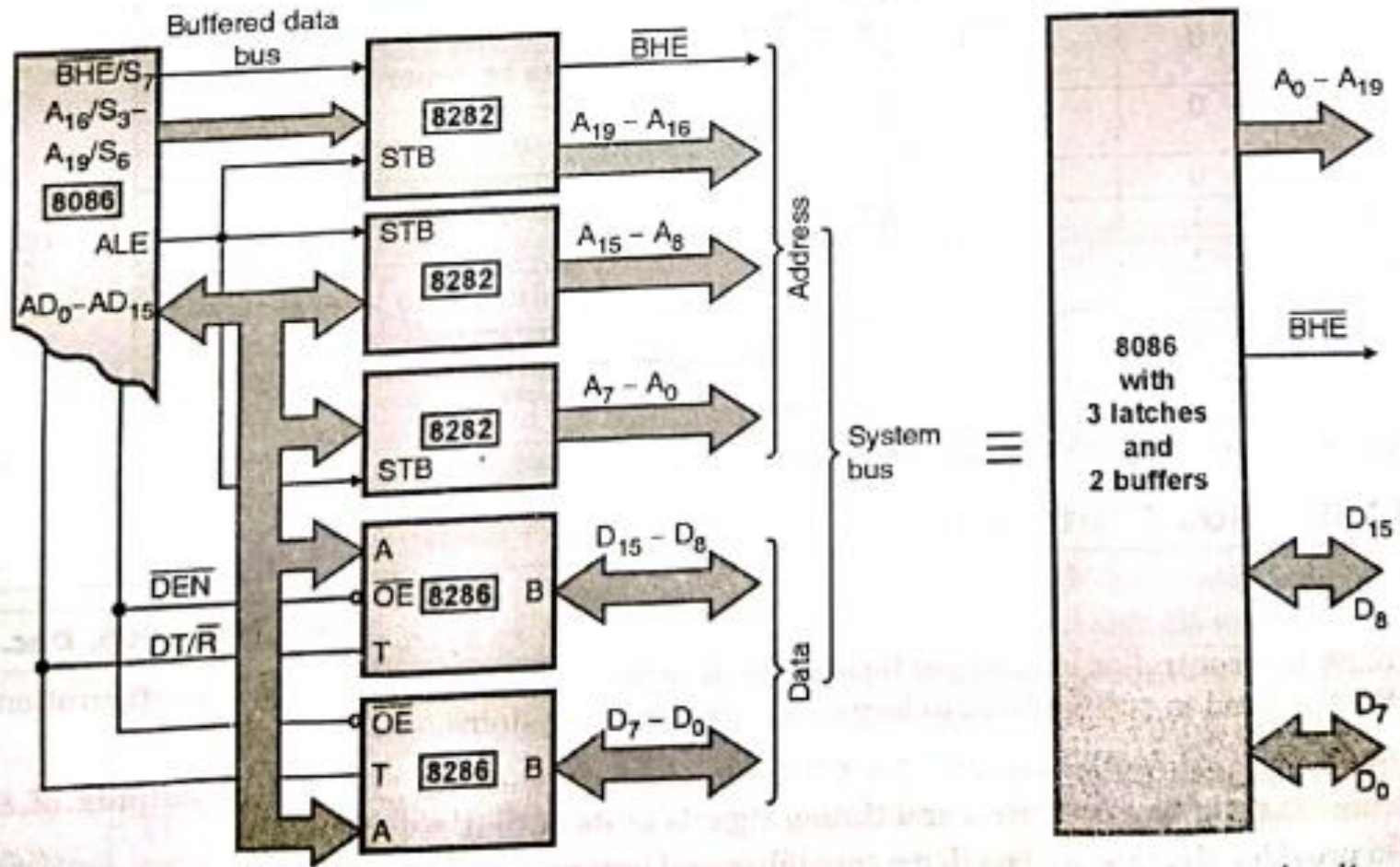


Fig. : Interfacing of transceiver 8286 with 8086

Demultiplexing Address and Data Bus



(a) Demultiplexing the address and data buses

(b) Simplified equivalent circuit

Control Bus

- The 8086/8088 processors have three buses namely the address bus, data bus and control bus.
- Let us now discuss about the last one i.e. the control bus.
- The 8086 provides three control lines namely $\overline{RD}$ (read), $\overline{WR}$ (write) and $M/\overline{IO}$ (Memory/input output port).
- We will use the decoder IC 74138 to convert these three lines into the following four signals.

- | | | |
|----|--------------|-------------------|
| 1. | Memory Read | $\overline{MEMR}$ |
| 2. | Memory Write | $\overline{MEMW}$ |
| 3. | I/O Read | $\overline{IOR}$ |
| 4. | I/O Write | $\overline{IOW}$ |

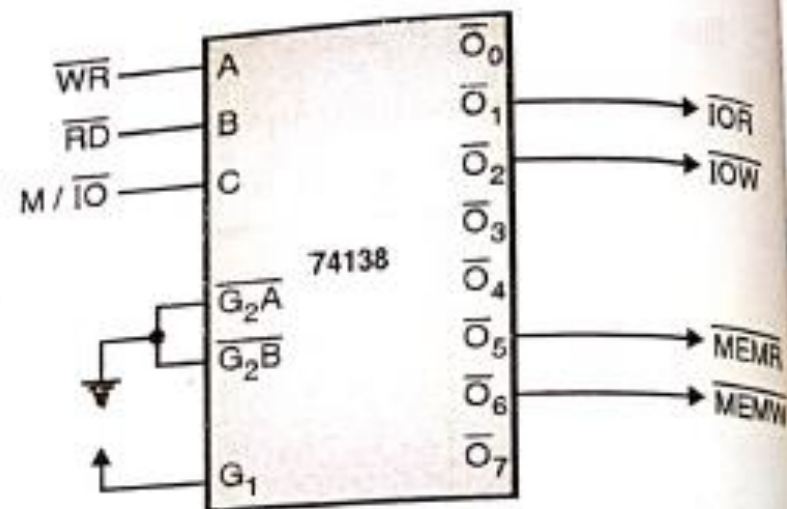


Fig. 3.4.4 : Generating control signals using 74LS138 decoder chip

Decoder-74138

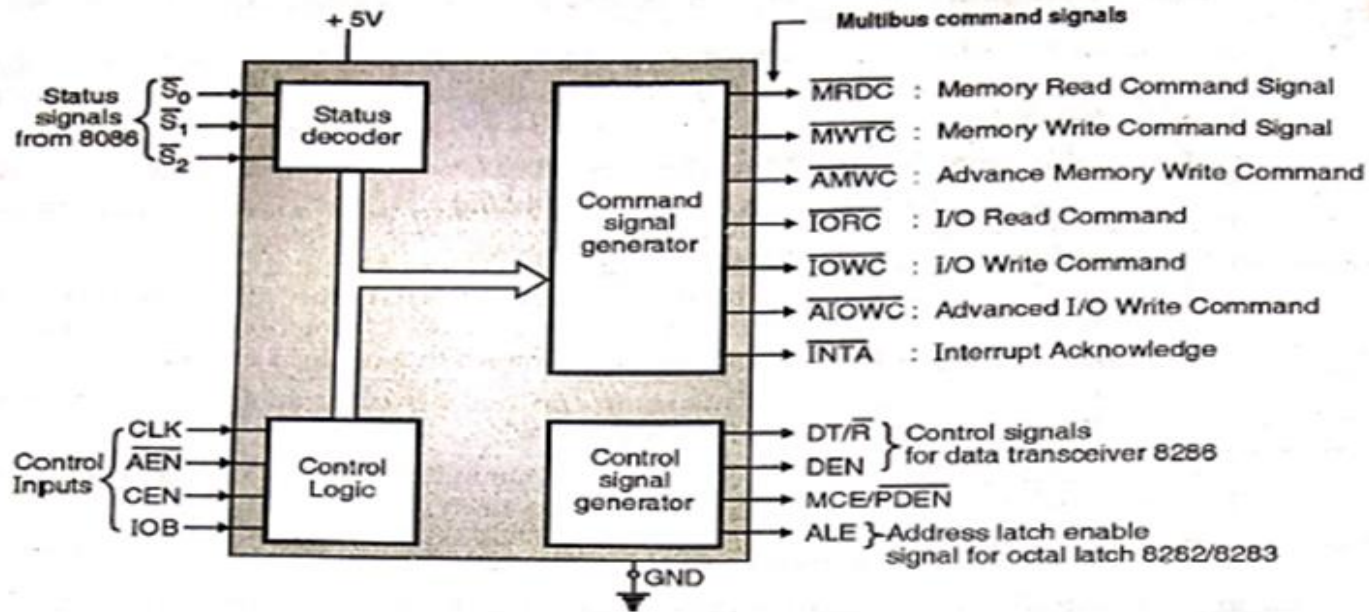
: Generating $\overline{\text{MEMR}}$, $\overline{\text{MEMW}}$, $\overline{\text{IOR}}$, $\overline{\text{IOW}}$

$\text{M}/\overline{\text{IO}}$	$\overline{\text{RD}}$	$\overline{\text{WR}}$	Operation
0	0	0	Invalid
0	0	1	I/O read ($\overline{\text{IOR}}$ to be generated)
0	1	0	I/O write ($\overline{\text{IOR}}$ to be generated)
0	1	1	No operation
1	0	0	Invalid
1	0	1	Memory read ($\overline{\text{MEMR}}$ to be generated)
1	1	0	Memory write ($\overline{\text{MEMW}}$ to be generated)
1	1	1	No operation

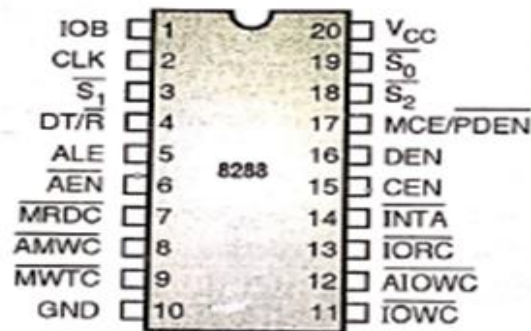
so decoder 74138 to generate control signals

8288- Bus Controller

- Used in maximum mode configuration of 8086.
- 8288 accepts the CLK signal along with So ,S1, S2 outputs of 8086 and generates the command , control and timing signals at its output.
- It also provides the bipolar bus drive capability and optimizes the system performance.



Simplified internal block diagram of 8288 bus controller



Pin diagram of 8288 bus controller

Pin Definitions

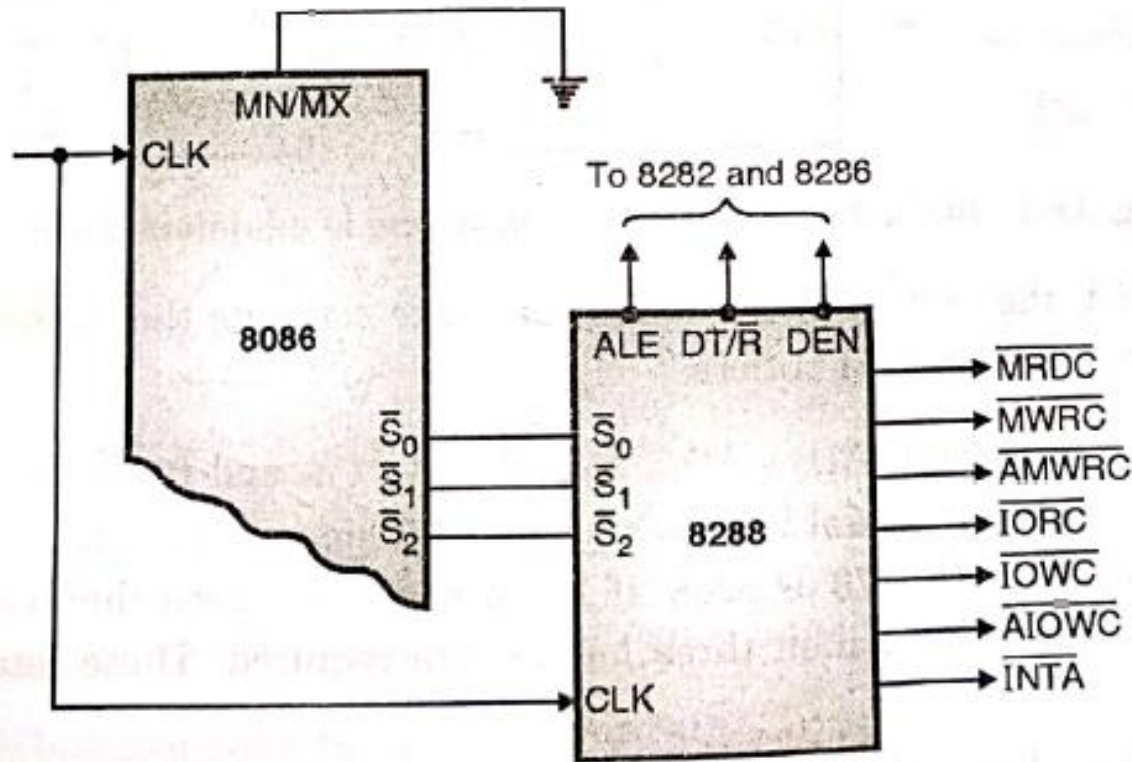
Name	I/O	Function
V _{CC}		+5V supply.
GND		Ground
$\bar{S}_0, \bar{S}_1, \bar{S}_2$	I	Status Input Pins : These pins are the status input pins from the 8086/8088 processors. The 8288 decodes these inputs to generate command and control signals at the appropriate time. When these pins are not in use (passive) they are all HIGH.
CLK	I	Clock : This is a clock signal from the 8284 clock generator and serves to establish synchronization when command and control signals are generated.
ALE	O	Address Latch Enable : This signal serves to strobe an address into the address latches. This signal is active HIGH and latching occurs on the falling (HIGH to LOW) transition. ALE is intended for use with transparent D type latches.
DEN	O	Data Enable : This signal serves to enable data transceivers onto either the local or system data bus. This signal is active HIGH.
DTR	O	Data Transmit/Receive : This signal establishes the direction of data flow through the transceivers. A HIGH on this line indicates Transmit (write to I/O or memory) and a LOW indicates Receive (read).

Name	I/O	Function
$\overline{\text{AEN}}$	I	Address Enable : $\overline{\text{AEN}}$ enables command outputs of the 8288. $\overline{\text{AEN}}$ does not affect the I/O command lines if the 8288 is in the I/O Bus mode (IOB tied HIGH).
$\overline{\text{CEN}}$	I	Command Enable : When this signal is LOW all 8288 command outputs and the $\overline{\text{DEN}}$ & $\overline{\text{PDEN}}$ control outputs are forced to their inactive T-state. When this signal is HIGH, these outputs are enabled.
$\overline{\text{IOB}}$	I	I/O Bus Mode : When the IOB is strapped HIGH the 8288 functions in the I/O Bus mode. When it is strapped LOW, the 8288 functions in the System Bus mode. (See sections on I/O Bus and System Bus modes i.e. 3.5.3).
$\overline{\text{IOWC}}$	O	I/O Write Command : This command line instructs an I/O device to read from the data bus. This signal is active LOW.
$\overline{\text{IORC}}$	O	I/O Read Command : This command line instructs an I/O device to drive its data onto the data bus. This signal is active LOW.
$\overline{\text{AIOWC}}$	O	Advanced I/O Write Command : The $\overline{\text{AIOWC}}$ issues an I/O Write command earlier in the machine cycle to give I/O devices an early indication of a write instruction. Its timing is the same as a read command signal. $\overline{\text{AIOWC}}$ is active LOW.
$\overline{\text{MWTC}}$	O	Memory Write Command : This command line instructs the memory to record the data present on the data bus. This signal is active LOW.
$\overline{\text{MRDC}}$	O	Memory Read Command : This command line instructs the memory to drive its data onto the data bus. This signal is active LOW.

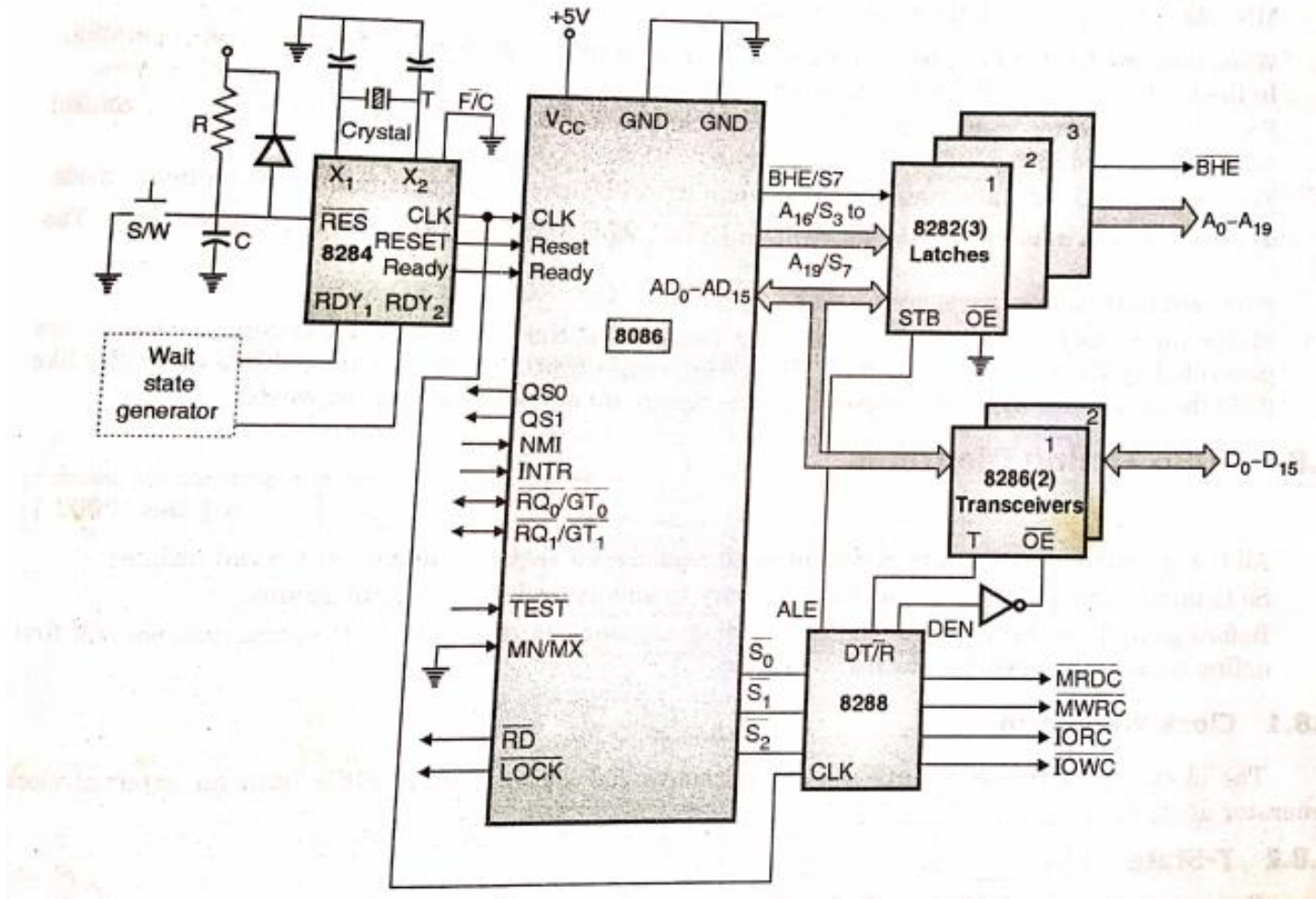
$\overline{\text{AMWC}}$	O	Advanced Memory Write Command : The $\overline{\text{AMWC}}$ issues a memory write command earlier in the machine cycle to give memory devices an early indication of a write instruction. Its timing is the same as a read command signal. $\overline{\text{AMWC}}$ is active LOW.
$\overline{\text{INTA}}$	O	Interrupt Acknowledge : This command line tells an interrupting device that its interrupt has been acknowledged and that it should drive vectoring information onto the data bus. This signal is active LOW.
$\text{MCE} / \overline{\text{PDEN}}$	O	This is a dual function pin. MCE (when IOB is tied LOW) : Master Cascade Enable occurs during an interrupt sequence and serves to read a Cascade Address from a master PIC (Priority Interrupt Controller) onto the data bus. The MCE signal is active HIGH. $\overline{\text{PDEN}}$ (when IOB is tied HIGH) : Peripheral Data Enable enables the data bus transceiver for the I/O bus during I/O instructions. It performs the same function for the I/O bus that DEN performs for the system bus. $\overline{\text{PDEN}}$ is active LOW.

Status of CPU			Status word (8086)	8288 command
$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$		
0	0	0	Interrupt acknowledge	$\overline{INTA}$
0	0	1	Read I/O port	$\overline{IORC}$: I/O read command
0	1	0	Write I/O port	$\overline{IOWC}$, $\overline{AIOWC}$: I/O or advanced I/O write command
0	1	1	Halt	None
1	0	0	Code Access	$\overline{MRDC}$: Memory read command
1	0	1	Read Memory	$\overline{MRDC}$: Memory read command
1	1	0	Write Memory	$\overline{MWTC}$, $\overline{AMWC}$: Memory or advanced memory write command
1	1	1	Passive	None

Interfacing 8288 with 8086



Interfacing of 8288 with 8086

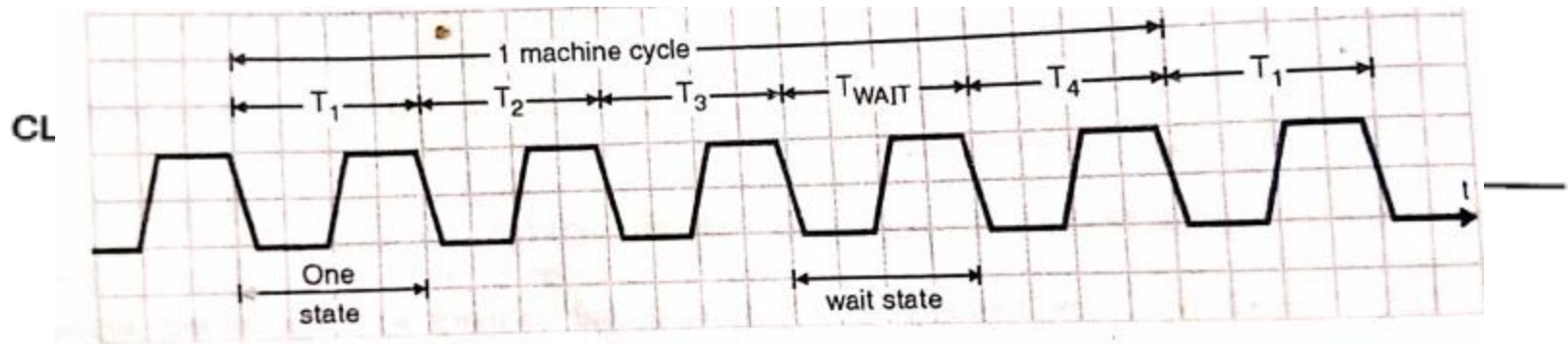


Timing Diagrams for 8086

All the operations in 8086 are carried out with a particular sequence in a synchronized manner, so it is necessary to understand the Timing Diagrams.

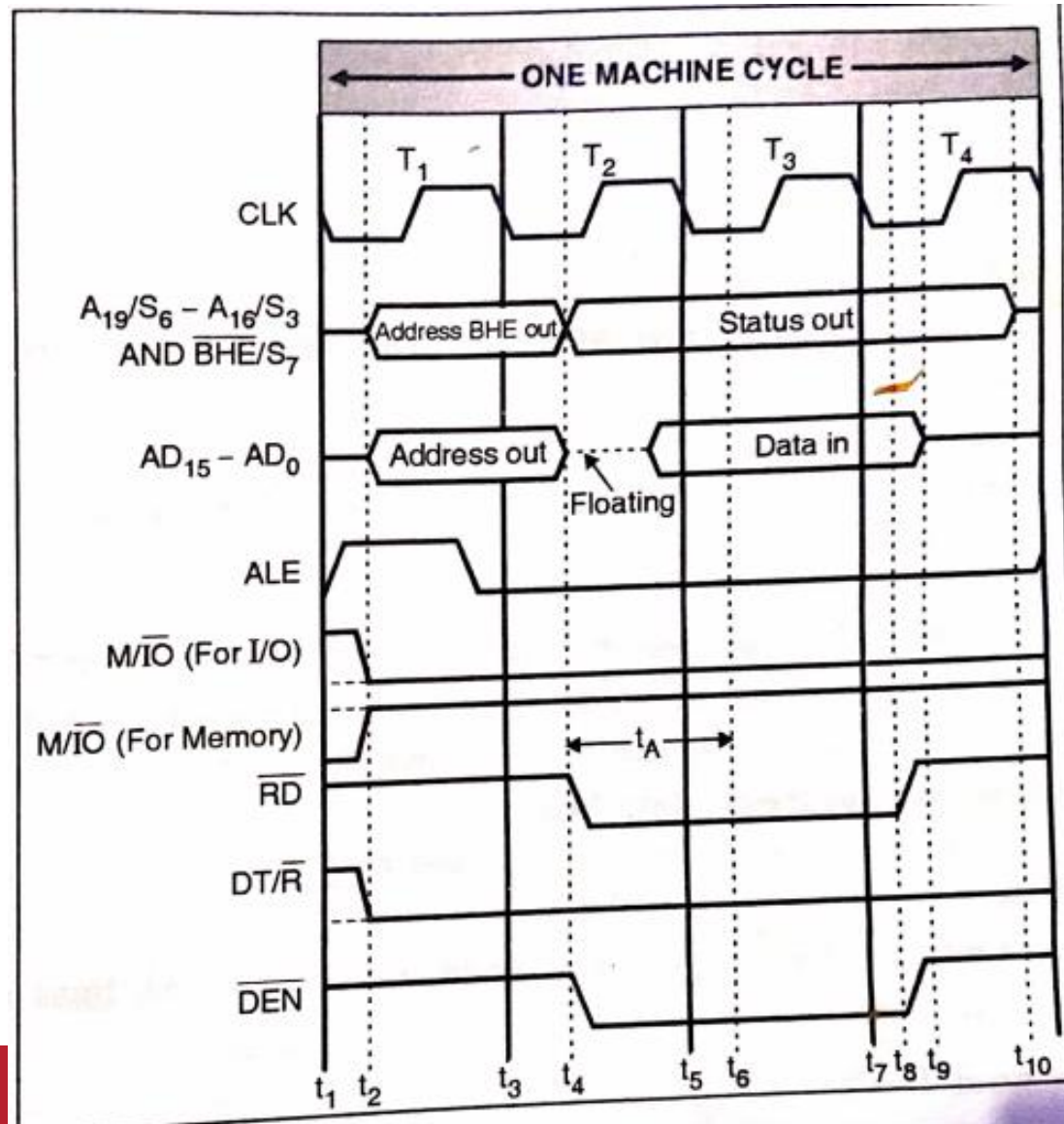
Clock Waveform: It represents the crystal controlled signal sent to 8086 from an external clock generator 8284.

Clock, T- State and Machine cycle

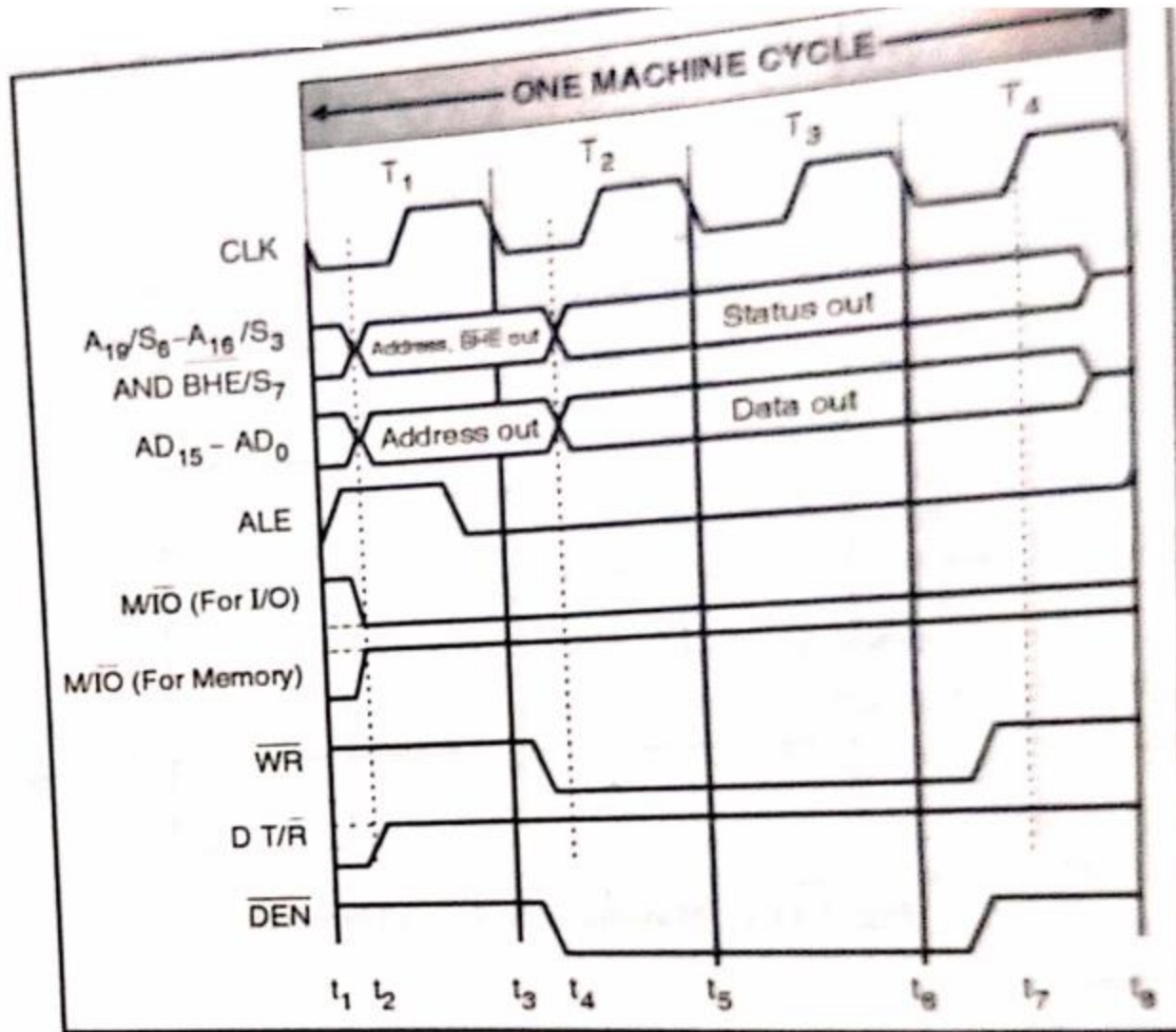


Clock, T-state, machine cycle etc.

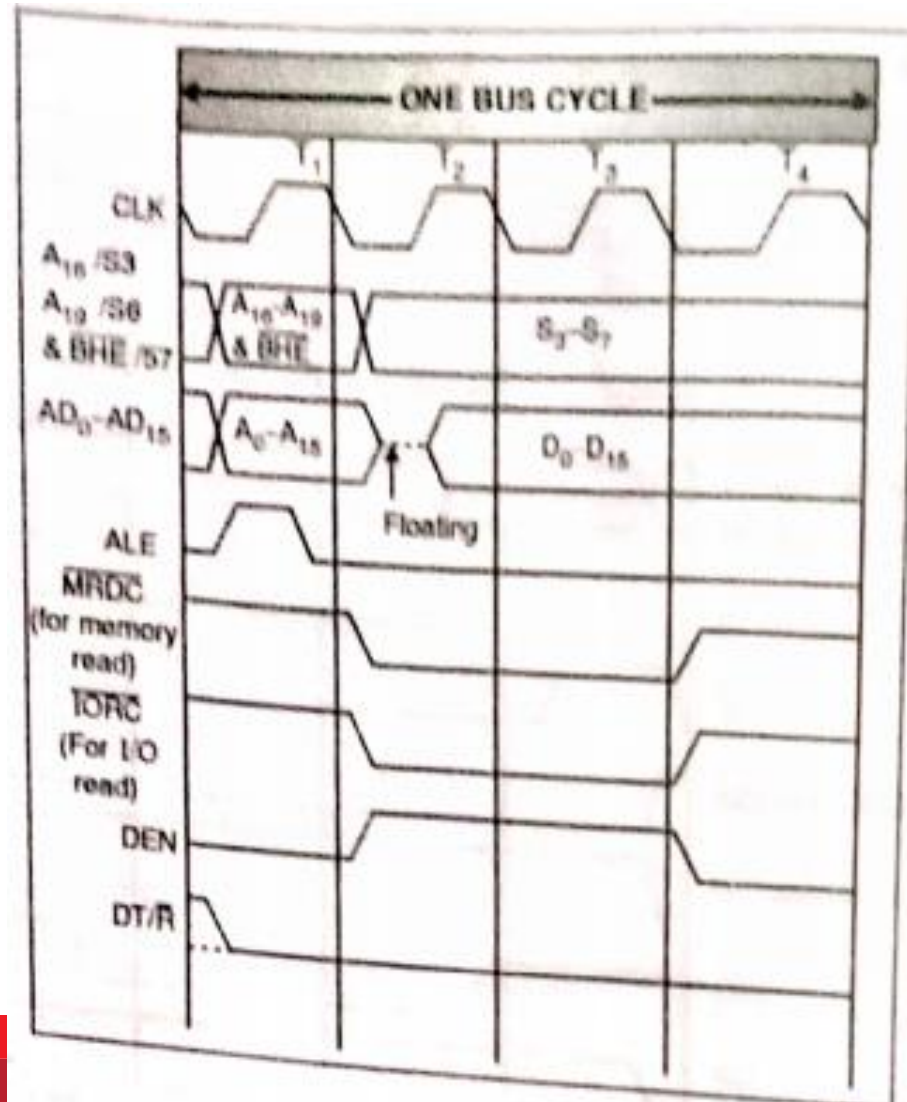
Timing diagram for Read machine cycle



Timing diagram for Write machine cycle



Maximum Mode-Read machine cycle



Maximum Mode-Write machine cycle

